

Code-Layout, Readability and Reusability

Date	3 July 2026
Team ID	XXXXXX
Project Name	Rising Waters – Flood Prediction System
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Proper indentation maintained in Python, HTML, CSS, and JavaScript files.
2	Proper File Structure	Files and folders are logically organized	Yes	Project organized into templates, static, model, dataset, and application files.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive variable names improve
4	Function / Method Names	Functions are descriptively named	Yes	Functions clearly indicate their purpose (e.g., prediction, preprocessing).
5	Code Comments	Inline and block comments explain logic	Partial	Comments are added where necessary but can be improved further.
6	Modular Design	Code is split into reusable functions/modules	Yes	ML model, preprocessing, and Flask application are organized into separate modules.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Unused and duplicate code has been minimized.
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic input validation and error handling are implemented.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Flask Application (app.py)	Python,Flask	Handles routing and prediction requests	High

2	Prediction Model	Scikit-learn (Decision Tree)	Used for all flood predictions	High
3	Data Preprocessing	Python, Pandas	Used during training and prediction	High
4	HTML Templates	HTML, CSS, JavaScript	Used across multiple web pages	Medium
5	Trained Model (.pkl)	Joblib/Pickle	Loaded whenever predictions are requested	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure with separate frontend, backend, and model files.
Readability	5	Code is easy to understand with meaningful names and consistent formatting.
Reusability	5	Modular design allows components to be reused across the project.
Documentation / Comments	4	Sufficient comments are included; additional documentation could further improve maintainability.
Overall Score	4.8/5	The project follows good coding practices and is maintainable and reusable.